

Varactor Corner Characterization Flow

1. Testbench 1: V_{dc} Sweep Characterization

Testbench: Tese_Varactor_characterization_tb1

Corners: TT25 (25°C, Typical), SS125 (125°C, Slow), FF-40 (−40°C, Fast)

Sweep: Parametric sweep of V_{dc} from 0.1 V to 0.6 V (30 linear points).

1.1 Minimum Frequency ($f_{min} = 6$ GHz) Analysis

OUTPUT	EXPRESSION
I_{dev} (Device_current_M7)	i(/V24/PLUS)
C_{meas}	abs(imag(Device_current_M7) / (2 * 3.14159 * 6e9))
C_{cont}	flatten(Cap_measuredAt1Mhz_M7, Vdc)
$\frac{dC}{dV}$	derivative(Cap_measuredAt1Mhz_M7_cont)
$V_{dc,opt}$	x_at_ymin(o(Deriv_Cap_measuredAt1Mhz_M7, analysisName="ac"), 0.1, 0.6, 1)

1.2 Maximum Frequency ($f_{max} = 9.5$ GHz) Analysis

OUTPUT	EXPRESSION
I_{dev} (Device_current_M7)	i(/V24/PLUS)
C_{meas}	abs(imag(Device_current_M7) / (2 * 3.14159 * 9.5e9))
C_{cont}	flatten(Cap_measuredAt1Mhz_M7, Vdc)
$\frac{dC}{dV}$	derivative(Cap_measuredAt1Mhz_M7_cont)
$V_{dc,opt}$	x_at_ymin(o(Deriv_Cap_measuredAt1Mhz_M7, analysisName="ac"), 0.1, 0.6, 1)

2. Testbench 2: Temperature Sweep Characterization

Testbench: Tese_Varactor_characterization_tb2

Sweep: Temperature sweep (temp) from -40°C to 125°C (11 points, step 16.5).

2.1 Minimum Frequency ($f_{\min} = 6\text{ GHz}$) Analysis

OUTPUT	EXPRESSION
I_{dev} (Device_current_M7)	i(/V24/PLUS)
Q-factor	abs(imag(Device_current_M7) / real(Device_current_M7))
C_{meas}	abs(imag(Device_current_M7) / (2 * 3.14159 * 6e9))

2.2 Maximum Frequency ($f_{\max} = 9.5\text{ GHz}$) Analysis

OUTPUT	EXPRESSION
I_{dev} (Device_current_M7)	i(/V24/PLUS)
Q-factor	abs(imag(Device_current_M7) / real(Device_current_M7))
C_{meas}	abs(imag(Device_current_M7) / (2 * 3.14159 * 9.5e9))